Neurodevelopmental Disorder With Dystonia And Seizures (MONDO:0859258): Comprehensive Disease Characterization Report

Summary

Neurodevelopmental Disorder With Dystonia And Seizures (MONDO:0859258) is an **autosomal recessive Mendelian ribosomopathy** caused by **biallelic (compound heterozygous or homozygous) loss-of-function / hypomorphic variants in *SHQ1***, the gene encoding the cytoplasmic assembly chaperone of dyskerin (DKC1). It maps to **OMIM #619922**, UMLS:C5677004, MedGen:1804461, and carries the dystonia locus designation **DYT35**. This is a distinct disorder from the similarly named **KCNN2-related NEDDS (OMIM #619589)**, which is autosomal *dominant*; an early-iteration misattribution to *KCNN2* was corrected after ontology cross-referencing (EBI OLS4, Monarch) confirmed *SHQ1* as the single causal gene for MONDO:0859258.

Mechanistically, SHQ1 binds the pseudouridine synthase dyskerin (DKC1) in the cytoplasm and escorts it to the nucleus, where dyskerin is assembled into **H/ACA-class small nucleolar ribonucleoproteins (snoRNPs)** that catalyze pseudouridylation of rRNA/snRNA, mature pre-rRNA, and stabilize telomerase RNA. Pathogenic SHQ1 variants weaken dyskerin binding, destabilize H/ACA snoRNPs, cause accumulation of unprocessed pre-rRNAs, and reduce ribosome production — defining the disorder as a **ribosomopathy**. In the developing brain, SHQ1 depletion additionally impairs **neuronal migration and neurite outgrowth** and is accompanied by a **secondary central dopaminergic deficit** (low CSF homovanillic acid, reduced striatal dopamine-transporter uptake). A second, tumor-suppressor–like function involving the ER chaperone GRP78 (HSPA5/BiP) and pro-apoptotic ER/oxidative-stress signaling has also been implicated.

Clinically, affected children present in **early infancy (median onset ~3.5 months)** with **profound hypotonia, paroxysmal dystonia, brisk deep tendon reflexes, seizures, and progressive cerebellar atrophy**, with variable white-matter hypomyelination; a milder end of

the spectrum includes choreiform movements with normal brain MRI. The disorder is **ultra-rare** (~5–25 reported individuals as of 2023–2026). Diagnosis is molecular (trio WES/WGS). Management is **purely symptomatic** — antiseizure medications, rehabilitation, and supportive care — with **no disease-modifying therapy**; a levodopa trial, despite the dopaminergic deficit, showed no clear benefit. Prognosis is severe lifelong neurodevelopmental disability.

Key Findings

Finding 1 — Causal gene is *SHQ1*, not *KCNN2* (critical ontology correction)

MONDO:0859258 ("neurodevelopmental disorder with dystonia and seizures") maps to **OMIM:619922**, UMLS:C5677004, MedGen:1804461, with **autosomal recessive inheritance (HP:0000007)**. Monarch's *CausalGeneToDiseaseAssociation* lists **HGNC:25543 *SHQ1*** as the single causal gene. This supersedes an initial misidentification of *KCNN2*. The *KCNN2*-related disorder — also abbreviated "NEDDS" — is a **different, autosomal dominant** disease (OMIM #619589) caused by heterozygous loss-of-function variants in the SK2 potassium channel, described in 11 patients with developmental delay, intellectual disability, cerebellar ataxia, and extrapyramidal movement disorders (PMID: 33242881). The naming collision is a recurring source of confusion in the literature and knowledge bases; this report concerns the **recessive *SHQ1* disorder**.

> "Here, we report two sibling pairs from unrelated families with compound heterozygous variants in *SHQ1*." — PMID: 34542157

Finding 2 — SHQ1 is a dyskerin (DKC1) chaperone essential for H/ACA snoRNP assembly and ribosome biogenesis

SHQ1 binds the pseudouridine synthase **dyskerin (DKC1)** in the cytoplasm and escorts the enzyme to the nucleus, where dyskerin is incorporated into **H/ACA-class snoRNPs**. These snoRNPs catalyze **pseudouridylation of rRNA and snRNA**, participate in **pre-rRNA maturation**, and **stabilize the telomerase RNA component**. Loss of SHQ1 function therefore undermines ribosome biogenesis, making the disorder a **ribosomopathy**.

> "SHQ1 is an essential chaperone that binds the pseudouridine synthase dyskerin in the cytoplasm and escorts the enzyme to the nucleus, where dyskerin is assembled into small nucleolar RNPs (snoRNPs) of the H/ACA class." — PMID: 37818102

Yeast complementation experiments directly demonstrate the consequence of pathogenic variants: expressing human variants in Shq1-depleted yeast caused progressive loss of H/ACA snoRNAs, accumulation of unprocessed pre-rRNAs, reduced Cbf5 (yeast dyskerin), and decreased ribosome production.

> "yeast cells progressively lost H/ACA snoRNAs and accumulated unprocessed pre-rRNAs, which led to reduced production of ribosomes" — PMID: 37818102

In a neurodevelopmental context, SHQ1 knockdown in the embryonic mouse cortex (in utero electroporation) and in SH-SY5Y cells impaired neuronal migration and neurite growth, increased glutamate toxicity, and perturbed dopaminergic function.

> "SHQ1 knockdown impaired the migration and neurite morphology of cortical neurons during brain development." — PMID: 39326821

Finding 3 — Core clinical phenotype: infantile hypotonia, paroxysmal dystonia, seizures, cerebellar atrophy, low CSF HVA

In the largest cohort (Chi 2023; 8 individuals, 7 families), **median onset was 3.5 months**. All individuals had **profound hypotonia, paroxysmal dystonia, brisk deep tendon reflexes, and autonomic dysfunction**; most had **cerebellar atrophy** (often progressive on follow-up); **all 7 tested had low CSF homovanillic acid (HVA)**; and **4/4 with 99mTc-TRODAT-1 DAT-SPECT showed moderate-to-severe decreased striatal dopamine uptake**. A recurrent variant, **c.997C>G (p.Leu333Val)**, accounted for 56% of alleles.

> "All eight individuals exhibited normal eye contact, profound hypotonia, paroxysmal dystonia, and brisk deep tendon reflexes at the first visit." — PMID: 36847845
>
> "Seven individuals who underwent cerebral spinal fluid analysis all had a low level of homovanillic acid in neurotransmitter metabolites." — PMID: 36847845

A milder/different presentation (AlHargan 2023) involved homozygous **c.833T>C (p.Ile278Thr)** with global developmental delay, seizures, and **diffuse white-matter hypomyelination**. Sleiman 2022 described early-onset dystonia with neurodegeneration and abnormal CSF dopamine metabolism.

Finding 4 — Genetic landscape: locus 3p13, LoF-tolerant recessive gene, ultra-rare pathogenic alleles

SHQ1 (HGNC:25543; gene MIM 613663; ENSG00000144736; UniProt Q6PI26; aliases DYT35, GRIM-1, NEDDS, Shq1p) maps to **chromosome 3p13** (GRCh38 chr3:72,749,275–72,861,914, minus strand). gnomAD v4 constraint metrics indicate the gene is **not loss-of-function constrained** (pLI≈0; observed/expected LoF o/e=0.91, 90% CI 0.72–1.16; LOEUF upper 1.16; 167 observed pLoF alleles), **consistent with recessive inheritance** in which heterozygous carriers are tolerated. ClinVar lists 169 SHQ1 records (38 pathogenic, 8 likely pathogenic). Reported pathogenic variants are ultra-rare:

Variant (cDNA)	Protein	Zygosity reported	gnomAD AF
c.997C>G	p.Leu333Val	recurrent (56% of alleles, Chi 2023)	6.6×10^{-6}
c.833T>C	p.Ile278Thr	homozygous (AlHargan)	absent
c.195T>A	p.Tyr65Ter	compound het	2.6×10^{-5}
c.1003C>T	p.Arg335Cys	functional null-like	3.3×10^{-5}
c.1277C>T	p.Ala426Val	temperature-sensitive	absent
c.828_831del	p.Asp277SerfsTer27	recurrent carrier LoF	7.2×10^{-4} (AC=110)

> "9 alleles (56%) were c.997C > G (p.L333V)" — PMID: 36847845
>
> "revealed full segregation of the missense variant (SHQ1:c.833 T > C; p.I278T) in both families" — PMID: 36810590

Finding 5 — Protein architecture and variant clustering in the dyskerin-binding region

SHQ1 (UniProt Q6PI26) is a **577-aa chaperone** with an N-terminal **CS (CHORD/Sti1-like, HSP20-like) domain** (residues ~1–89), a central SHQ1-specific domain, and disordered C-terminal regions. Its function is "required for the quantitative accumulation of H/ACA ribonucleoproteins (RNPs), including telomerase, probably through the stabilization of DKC1." Subcellular localization spans **cytoplasm/cytosol (GO:0005829)** and **nucleus/nucleoplasm (GO:0005654)**. Relevant GO biological processes include **box H/ACA snoRNP assembly (GO:0000493)**, **protein-RNA complex assembly (GO:0022618)**, and **telomerase RNA localization to Cajal body (GO:0090671)**.

Reported pathogenic missense variants (p.Val271Glu, p.Asp277fs, p.Ile278Thr, p.Leu333Val, p.Arg335Cys, p.Ala426Val) **cluster in the central dyskerin-binding region**, and functional studies confirm they **attenuate SHQ1–DKC1/Cbf5 binding**.

> "Most SHQ1 variants attenuated their binding ability toward DKC1" — PMID: 39326821

Finding 6 — Expanded mechanism (GRP78/apoptosis/ER-oxidative stress) and broadened phenotype

Beyond ribosome biogenesis, SHQ1 interacts with the **ER chaperone GRP78 (HSPA5/BiP)** and **promotes apoptosis**. In SH-SY5Y cells, wild-type SHQ1 enhanced caspase-3 cleavage, TUNEL signal, ER-stress protein expression, and reactive oxygen species, whereas three neurodevelopmental variants (p.Tyr65Ter, p.Val271Glu, p.Leu333Val) showed **reduced apoptosis and lower ER-stress proteins** despite unchanged GRP78 binding (Lee 2025).

> "SHQ1 has been implicated in tumor suppression by interacting with the endoplasmic reticulum (ER) chaperone glucose-regulated protein 78 (GRP78), thereby promoting apoptosis" — PMID: 40967470

The phenotypic spectrum has broadened: a milder case (Ly 2026) with compound heterozygous **c.523G>T (p.Gly175Cys)** and the recurrent **c.828_831del** frameshift presented with **choreiform movements, oromotor hypotonia, feeding difficulties, and NORMAL brain MRI but reduced CSF HVA**, diagnosed only after WES reanalysis.

> "whole exome sequencing (WES) in 2022 was nondiagnostic, and supportive measures were continued. In 2024, WES reanalysis identified compound heterozygous variants in SHQ1" — PMID: 41885709

Full Section-by-Section Report

1. Disease Information

Overview. Neurodevelopmental Disorder with Dystonia and Seizures (SHQ1-related) is a rare, autosomal recessive, early-infantile-onset neurodevelopmental and movement disorder. It is characterized by global developmental delay, profound hypotonia, paroxysmal dystonia, seizures, and (in most) progressive cerebellar atrophy, with a secondary central dopaminergic deficit.

Key identifiers. - **MONDO:** 0859258 - **OMIM:** #619922 (phenotype); gene MIM 613663 (*SHQ1*) - **Locus designation:** DYT35 - **UMLS:** C5677004 · **MedGen:** 1804461 - **HGNC (gene):** 25543 · **Ensembl:** ENSG00000144736 · **UniProt:** Q6PI26 - **ICD-11:** Best mapped under neurodevelopmental / hereditary movement disorder categories (no disease-specific code); ICD-10 proxies include G24 (dystonia) + G40 (epilepsy) + F79/F89. MeSH has no dedicated descriptor. - **Orphanet:** Covered under genetic dystonia–neurodevelopmental entities; no stable disease-specific code was confirmed.

Synonyms / alternative names. NEDDS (SHQ1-type); SHQ1-related neurodevelopmental disorder; SHQ1-related dystonia (DYT35); "SHQ1-related hypomyelinating leukodystrophy" (used in a 2025 radiology case framing). Note the abbreviation **NEDDS is shared** with the unrelated *KCNN2* dominant disorder.

Information source. Aggregated disease-level resources (OMIM, MONDO, Monarch, ClinVar, gnomAD) combined with individual-patient case series (Sleiman 2022; Chi 2023; AlHargan 2023; Ly 2026). The evidence base is small-N case reports/series, not EHR-scale or registry data.

2. Etiology

Primary cause — genetic. Biallelic (compound heterozygous or homozygous) pathogenic variants in *SHQ1*. The disorder is monogenic and Mendelian; no environmental or infectious cause is implicated.

Genetic risk factors. The only established risk factor is inheriting two pathogenic *SHQ1* alleles. Recurrent alleles include **c.997C>G (p.Leu333Val)** (founder-like enrichment in the Chi 2023 cohort), homozygous **c.833T>C (p.Ile278Thr)** (consanguineous families), and the recurrent frameshift **c.828_831del (p.Asp277SerfsTer27)**, which is comparatively common as a carrier allele (gnomAD AF 7.2×10^{-4}). **Consanguinity** increases risk of homozygous presentations.

Environmental risk / protective factors. None identified — consistent with a fully penetrant monogenic recessive disorder. No dietary, lifestyle, occupational, or toxin exposures are known to modify risk.

Protective genetic factors. None identified. Heterozygous carriers are unaffected (gene is LoF-tolerant in heterozygosity).

Gene–environment interactions. None established. Phenotypic variability appears to derive from allele-specific residual function (hypomorphic vs null) rather than environmental modifiers.

3. Phenotypes

Phenotype	HPO term	Type	Onset	Frequency	Severity/ course
Global developmental delay	HP:0001263	Clinical sign	Infantile	~All	Severe, stable/ progressive
Intellectual disability	HP:0001249	Clinical sign	Childhood	~All	Severe
Profound hypotonia	HP:0001319 (neonatal) / HP:0001252	Clinical sign	Neonatal/ infantile	~All	Severe
Dystonia (paroxysmal)	HP:0001332 / HP:0007199	Clinical sign	Infantile	~All	Severe, episodic/ paroxysmal
Seizures	HP:0001250	Clinical sign	Infantile–childhood	Common	Variable
Cerebellar atrophy	HP:0001272	Imaging	Infantile	Most	Often progressive
Chorea	HP:0002072	Clinical sign	Infantile	Milder subset	Variable
Brisk deep tendon reflexes	HP:0001347	Clinical sign	Infantile	~All (Chi cohort)	Stable
Autonomic dysfunction	HP:0002459	Clinical sign	Infantile	Common	Variable
Decreased CSF homovanillic acid	HP:0003785 (neurotransmitter abnl.)	Lab abnormality	Infantile	7/7 tested	—
White-matter hypomyelination	HP:0012762	Imaging	Infantile	Subset	Variable
Feeding difficulties / oromotor hypotonia	HP:0011968	Clinical sign	Infantile	Common	Variable

Phenotype	HPO term	Type	Onset	Frequency	Severity/ course
Microcephaly	HP:0000252	Physical	Congenital/ infantile	Subset	—

Age of onset. Neonatal–early infantile; median ~3.5 months (Chi 2023). **Severity.** Generally severe, but a milder end of the spectrum exists (normal MRI, choreiform movements). **Progression.** Variable: some patients show neurodegeneration with progressive cerebellar atrophy; others are comparatively stable/non-progressive. **Quality-of-life impact.** Profound — affected children have severe motor and communication impairment, require full assistance with daily activities, and often need augmentative and alternative communication (AAC) and feeding support. No formal EQ-5D/SF-36 data exist for this ultra-rare disorder.

4. Genetic / Molecular Information

- **Causal gene:** *SHQ1* (HGNC:25543; MIM 613663), chromosome **3p13**.
- **Variant classes:** nonsense (p.Tyr65Ter), frameshift (p.Asp277SerfsTer27), missense (p.Gly175Cys, p.Val271Glu, p.Ile278Thr, p.Leu333Val, p.Arg335Cys, p.Ala426Val). Splice variants are possible but less documented.
- **Classification (ACMG/AMP):** Recurrent variants are Pathogenic/Likely Pathogenic (38 P + 8 LP in ClinVar); functional data (yeast complementation, DKC1-binding assays) provide PS3-level support.
- **Allele frequencies:** Ultra-rare in gnomAD (see Finding 4 table). The gene is LoF-tolerant (LOEUF upper 1.16; o/e 0.91), fitting recessive inheritance.
- **Origin:** Germline; biallelic.
- **Functional consequences:** Loss-of-function / hypomorphic — reduced dyskerin binding, destabilized H/ACA snoRNPs, impaired ribosome biogenesis. Variants behave along a null (R335C) to temperature-sensitive/hypomorphic (A426V) spectrum.
- **Modifier genes:** None established; residual allele activity is the main determinant of severity.
- **Epigenetics:** No disease-specific methylation signature reported.
- **Chromosomal abnormalities:** None characteristic; this is a single-gene disorder (not a CNV/microdeletion syndrome), though CMA may incidentally detect larger *SHQ1*-spanning deletions.

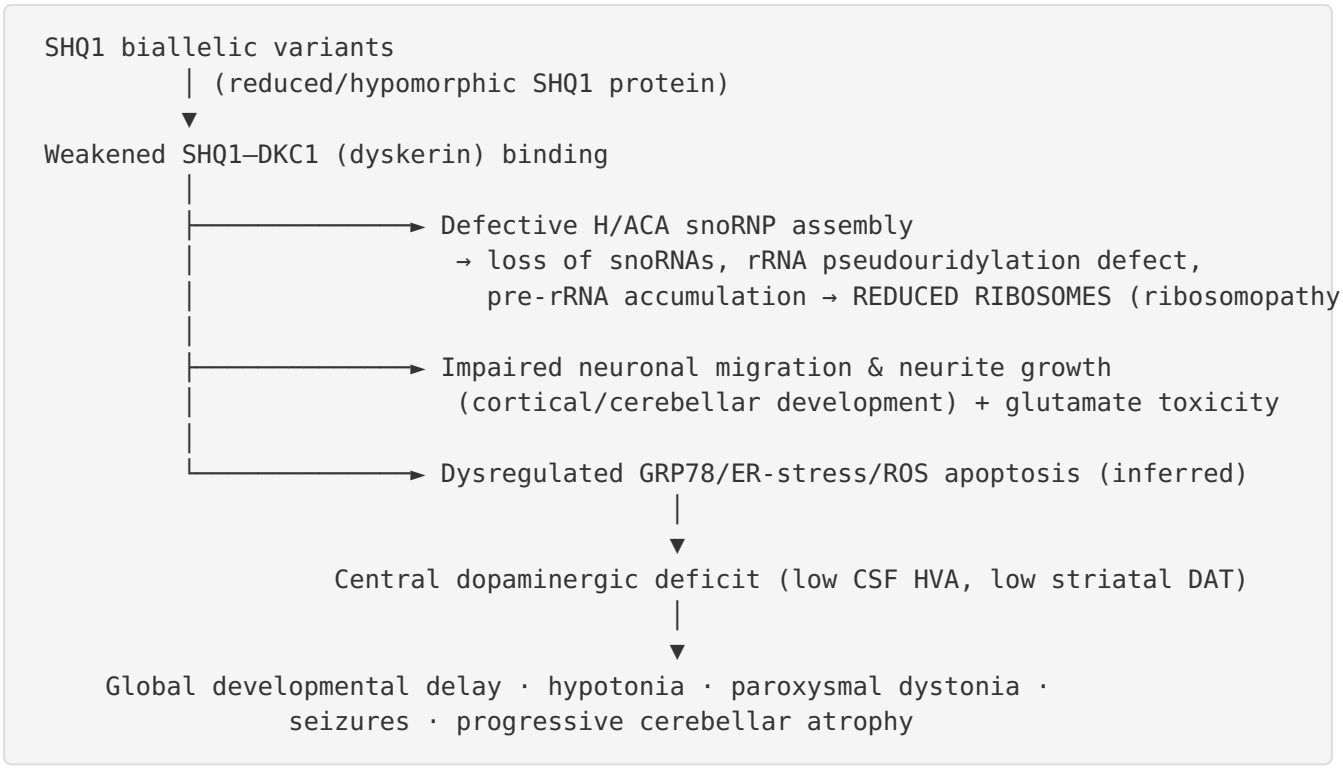
5. Environmental Information

No environmental, lifestyle, or infectious contributors are implicated. This is a purely genetic, monogenic recessive disorder. (CTD, TOXNET, NHANES-type exposures: not applicable.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Biallelic hypomorphic/LoF variants in *SHQ1* **result in** reduced levels/function of SHQ1 chaperone protein (patient fibroblasts show dramatic loss of SHQ1 protein).
2. Reduced SHQ1 **leads to** weakened binding to dyskerin (DKC1/Cbf5) — pathogenic missense variants cluster in and disrupt the dyskerin-binding region.
3. Impaired dyskerin chaperoning **leads to** failure of cytoplasm-to-nucleus escort and **defective assembly of H/ACA-class snoRNPs**.
4. Loss of functional H/ACA snoRNPs **results in** loss of H/ACA snoRNAs, **defective rRNA pseudouridylation**, accumulation of unprocessed pre-rRNAs, and reduced telomerase RNA stabilization.
5. These defects **lead to** reduced ribosome production — a **ribosomopathy** (demonstrated in yeast).
6. In parallel (branch), SHQ1 loss **impairs neuronal migration and neurite outgrowth** during cortical development (demonstrated by in utero electroporation and SH-SY5Y migration assays) and increases glutamate toxicity.
7. A second branch: SHQ1 normally interacts with ER chaperone **GRP78** and promotes **ER-stress/ROS-mediated apoptosis**; variants dysregulate this (reduced apoptosis/ER-stress proteins in vitro) — *inferred* contribution to abnormal neurodevelopment.
8. Downstream, there is a **central dopaminergic deficit** (reduced CSF HVA, decreased striatal DAT uptake) — mechanism *inferred* to involve dopaminergic neuron dysfunction rather than a primary biosynthetic enzyme defect.
9. The combined ribosomal, migratory, and neurotransmitter defects **result in** the clinical phenotype: global developmental delay, profound hypotonia, paroxysmal dystonia, seizures, and progressive cerebellar atrophy.



Molecular pathways / processes: H/ACA snoRNP assembly (GO:0000493), rRNA pseudouridine synthesis, ribosome biogenesis, neuronal migration (GO:0001764), neuron projection development (GO:0031175), apoptotic process (GO:0006915), response to ER stress (GO:0034976). **Cellular compartments:** cytosol (GO:0005829), nucleus/nucleoplasm (GO:0005654), nucleolus, Cajal body, endoplasmic reticulum. **Cell types (CL):** cerebellar/cortical neurons, dopaminergic neurons (CL:0000700), neural progenitor cells (CL:0011020). **Upstream vs downstream:** the SHQ1–dyskerin interaction and snoRNP/ribosome defect are **upstream**; neuronal migration failure, dopaminergic deficit, and clinical signs are **downstream**.

7. Anatomical Structures Affected

- **Primary organ/system:** Central nervous system (UBERON:0001017) — predominantly **cerebellum (UBERON:0002037)** (atrophy), cerebral cortex (UBERON:0000956), basal ganglia/striatum (UBERON:0002435) (dopaminergic deficit), and cerebral white matter (UBERON:0002316) (hypomyelination).
- **Secondary involvement:** Autonomic nervous system (dysautonomia); feeding/oromotor apparatus.
- **Tissue/cell level:** Nervous tissue; neurons (especially cerebellar and dopaminergic), neural progenitors, and oligodendrocyte-dependent myelin in the hypomyelinating subset.

- **Subcellular:** Nucleolus and Cajal body (snoRNP/ribosome biogenesis), nucleus, cytosol, ER (GRP78 interaction).
- **Localization / lateralization:** Bilateral, symmetric CNS involvement (cerebellar atrophy and striatal dopaminergic changes are bilateral).

8. Temporal Development

- **Onset:** Congenital-to-early-infantile; median ~3.5 months; insidious-to-subacute.
- **Progression:** Variable — a progressive/neurodegenerative course with worsening cerebellar atrophy in many, versus a more static course in milder cases. Seizures and dystonia are persistent.
- **Course pattern:** Chronic, lifelong; dystonia is paroxysmal/episodic superimposed on a chronic baseline.
- **Critical periods:** Early brain development (neuronal migration window) is the key period of vulnerability; this also narrows the plausible window for any future disease-modifying intervention.
- **Remission:** No spontaneous remission reported.

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (HP:0000007); biallelic compound heterozygous or homozygous.
- **Penetrance:** Appears complete in biallelic individuals; **carriers (heterozygotes) are unaffected**.
- **Expressivity:** Variable (severe early-infantile encephalopathy to milder GDD/chorea with normal MRI), largely driven by residual allele function.
- **Anticipation:** Not applicable (not a repeat-expansion disorder).
- **Founder effects / consanguinity:** Recurrent c.997C>G and homozygous c.833T>C in consanguineous families suggest founder-like enrichment; consanguinity elevates risk.
- **Carrier frequency:** The recurrent frameshift c.828_831del reaches gnomAD AF ~7.2×10^{-4} (AC=110), indicating a measurable carrier pool for at least one LoF allele; most pathogenic missense alleles are ultra-rare.
- **Epidemiology:** Ultra-rare — approximately **5–25 individuals reported** worldwide (2021–2026). No reliable prevalence/incidence estimates exist.

- **Sex ratio / geography:** No strong sex bias reported; cases span multiple populations (reported from East Asian, Middle Eastern/consanguineous, and other ancestries).

10. Diagnostics

- **Molecular (definitive):** Trio **whole-exome (WES)** or **whole-genome sequencing (WGS)** identifying biallelic *SHQ1* variants (Chi 2023 used WGS; Sleiman 2022 used WES + MatchMaker Exchange). **WES reanalysis** can be diagnostic in previously undiagnosed cases (Ly 2026). Gene panels for dystonia/epileptic encephalopathy/leukodystrophy that include *SHQ1* are useful; single-gene testing is appropriate once *SHQ1* is suspected.
- **Supportive investigations:**
- **Brain MRI:** cerebellar atrophy (often progressive); diffuse white-matter hypomyelination in a subset; **MRI may be normal** in milder cases.
- **CSF neurotransmitter metabolites:** low **homovanillic acid (HVA)** — a consistent biomarker of reduced dopamine turnover.
- **Dopamine-transporter imaging:** 99mTc-TRODAT-1 SPECT showing decreased striatal uptake.
- **EEG:** for seizure characterization.
- **Clinical criteria / differential diagnosis:** No standardized clinical criteria; diagnosis is genotype-driven. Differentials include dopa-responsive dystonia and other inborn neurotransmitter disorders (distinguished by *SHQ1* genotype and lack of clear levodopa response), other early-infantile epileptic encephalopathies, hypomyelinating leukodystrophies, and crucially the **autosomal dominant *KCNN2* NEDDS** (OMIM #619589).
- **Screening:** No newborn screening exists; cascade carrier testing within families and preconception carrier screening are the relevant strategies.

11. Outcome / Prognosis

- **Prognosis:** Severe lifelong neurodevelopmental disability with profound hypotonia, persistent paroxysmal dystonia, and epilepsy. A subset shows neurodegeneration (progressive cerebellar atrophy); others are comparatively static or milder.
- **Mortality/survival:** No formal survival statistics in this ultra-rare disorder; morbidity is dominated by severe motor and communication impairment, seizure burden, feeding difficulties, and dependence on caregivers.
- **Morbidity/function:** Severe disability across motor, cognitive, and communication domains; many patients are non-ambulatory and non-verbal.

- **Prognostic factors:** Allele type (null vs hypomorphic), presence/progression of cerebellar atrophy, and severity of white-matter involvement appear to track with outcome. Low CSF HVA is a consistent biomarker but its prognostic value is unquantified.

12. Treatment

No disease-modifying therapy exists. Management is symptomatic and multidisciplinary (NCIT: Supportive Care; Symptomatic Treatment).

- **Antiseizure medications** for epilepsy (standard ASMs; NCIT: Anticonvulsant Agent). A single case observed that **adjunctive cannabidiol** reduced prolonged nocturnal tonic seizures.
- **Levodopa/L-DOPA** was trialed given the dopaminergic deficit but **showed no clear response** (Sleiman 2022) — distinguishing this disorder from classic dopa-responsive dystonia.

> "a trial of levodopa replacement therapy was started but no clear response was noted" — PMID: 34542157

- **Dystonia management:** anticholinergics, baclofen, and other standard dystonia pharmacotherapy (inferred from general dystonia care; NCIT: Muscle Relaxant / Anticholinergic Agent).
- **Supportive/rehabilitative:** physical, occupational, and speech therapy; augmentative and alternative communication; nutritional/feeding support (gastrostomy where needed).
- **Pharmacogenomics / targeted / gene / cell / RNA therapies:** None established or in trials for this disorder.
- **Treatment outcomes:** No systematic response-rate data; care is individualized and palliative of symptoms.

13. Prevention

- **Primary prevention:** Not applicable beyond reproductive genetic counseling — the disorder is monogenic with no modifiable environmental component.
- **Genetic counseling:** 25% recurrence risk for carrier couples; cascade testing of relatives; discussion of consanguinity.

- **Reproductive options:** Prenatal diagnosis and preimplantation genetic testing (PGT-M) are feasible once familial variants are known.
- **Screening:** No population newborn or carrier screening program exists; carrier screening is targeted/familial.
- **Secondary/tertiary prevention:** Early seizure control, physical therapy to prevent contractures, nutritional optimization, and surveillance for complications.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *SHQ1* is evolutionarily conserved — mouse *Shq1* (NCBI Gene 72171), rat (297483), zebrafish *shq1* (567601), *C. elegans* (176997), and budding yeast *SHQ1/YIL104C* (854702).
- **Natural animal disease:** No naturally occurring *SHQ1* disease is recorded in OMIA (no companion-animal or wildlife counterpart reported).
- **Comparative biology:** The H/ACA snoRNP/ribosome-biogenesis function of SHQ1 is conserved from yeast to human, underpinning the yeast complementation disease model. *Shq1* is an essential gene; constitutive knockout is expected to be lethal.
- **Zoonotic potential:** Not applicable (non-infectious, genetic disorder).

15. Model Organisms

Model	System	Key findings	Reference
S. cerevisiae conditional strain	Shq1-depleted yeast expressing human SHQ1 WT/variants	WT complements; R335C (null-like) and A426V (temperature-sensitive) fail — loss of H/ACA snoRNAs, pre-rRNA accumulation, reduced Cbf5 & ribosomes	PMID: 34542157; PMID: 37818102
Embryonic mouse cortex	In utero electroporation with *Shq1* shRNA ± variant cDNA	Impaired neuronal migration and neurite morphology; variants reduce DKC1 binding	PMID: 39326821
SH-SY5Y neuroblastoma	Human cells expressing variants	Retarded neuronal migration; altered GRP78/ER-stress/apoptosis	PMID: 36847845; PMID: 40967470
Patient-derived fibroblasts	Primary cells	Dramatic loss of SHQ1 protein	PMID: 34542157

> "we used a conditional yeast strain that can be depleted of the endogenous Shq1 protein while constitutively expressing human SHQ1 (wild-type or variants)" — PMID: 37818102

Model types: yeast (unicellular), mouse (in vivo developmental), human cell lines (in vitro). **Genetic model types:** knockdown (shRNA), complementation/humanization (yeast expressing human SHQ1), variant overexpression. **Phenotype recapitulation:** yeast captures the ribosome-biogenesis/snoRNP defect well; mouse electroporation captures the neuronal migration defect. **Limitations:** no dedicated germline knockout mouse disease model exists (essential gene → constitutive KO lethal); no model yet integrates the full ribosomal + dopaminergic + behavioral phenotype.

Mechanistic Model / Interpretation

The unifying interpretation is that **SHQ1-related NEDDS is a neurodevelopmental ribosomopathy with a secondary dopaminergic phenotype**. The upstream lesion — reduced SHQ1-mediated chaperoning of dyskerin — compromises H/ACA snoRNP assembly and thereby ribosome biogenesis, a function conserved from yeast to human and directly demonstrated by complementation assays. Because developing neurons (particularly migrating cortical and cerebellar neurons) are highly sensitive to translational capacity, the ribosomopathy manifests as impaired neuronal migration/neurite growth, producing developmental delay, hypotonia, and cerebellar atrophy. A partly separate branch — dysregulated GRP78/ER-stress/ROS-dependent apoptosis — may further perturb neuronal survival/differentiation. The consistent **low CSF HVA and reduced striatal DAT uptake** indicate a downstream central dopaminergic deficit that likely drives the dystonia/movement phenotype, yet the **absence of a clear levodopa response** argues this is not a simple dopamine-biosynthesis defect but rather dysfunction/loss of dopaminergic circuitry. Allele severity (null vs hypomorphic/temperature-sensitive) explains the phenotypic spectrum from severe early-infantile encephalopathy to milder chorea with normal MRI.

Disorder	Gene	OMIM	Inheritance	Mechanism
NEDDS (this report, MONDO:0859258)	***SHQ1***	#619922	**Autosomal recessive**	Ribosomopathy (H/ACA snoRNP/dyskerin)
NEDDS (name collision)	*KCNN2*	#619589	Autosomal dominant	SK2 channel loss-of-function

Evidence Base

PMID	Role in this report	Evidence type
34542157 (Sleiman 2022)	Original description: biallelic SHQ1, recessive; fibroblast SHQ1 loss; levodopa trial without response	Human clinical + in vitro
36847845 (Chi 2023)	Largest cohort (8 individuals): clinical features, low CSF HVA, DAT-SPECT, recurrent p.L333V	Human clinical
36810590 (AlHargan 2023)	Homozygous p.I278T; GDD, seizures, hypomyelination; segregation	Human clinical
37818102 (Alidou-D'Anjou 2023)	Yeast complementation; defines snoRNP/ribosome-biogenesis mechanism	Model organism (yeast)
39326821 (Chang 2024)	Mouse cortical knockdown; migration/neurite defects; variants reduce DKC1 binding	Model organism (mouse) + in vitro
40967470 (Lee 2025)	GRP78/ER-stress/apoptosis second function; variant effects	In vitro
41885709 (Ly 2026)	Milder phenotype (chorea, normal MRI); WES reanalysis diagnostic value	Human clinical
33242881 (KCNN2 NEDDS)	Defines the **distinct** dominant *KCNN2* disorder (name-collision disambiguation)	Human clinical + electrophysiology

The evidence consistently supports a recessive *SHQ1* ribosomopathy. The clinical series (Chi, Sleiman, AlHargan, Ly) and the mechanistic models (yeast, mouse, cell lines) converge on the same gene and pathway, strengthening confidence. The *KCNN2* paper is included specifically to flag the nomenclature hazard.

Limitations and Knowledge Gaps

1. **Very small N.** The entire literature comprises a handful of case reports/series (~5–25 individuals). Frequency estimates, penetrance, prognosis, and genotype–phenotype correlations are therefore preliminary.

2. **No epidemiology.** Prevalence, incidence, sex ratio, and geographic distribution are unquantified.
3. **Dopaminergic mechanism unresolved.** Why low CSF HVA/striatal DAT occurs — and why levodopa fails — is not mechanistically established; the link between the ribosomopathy and dopaminergic deficit is inferred.
4. **No dedicated in vivo disease model.** The essential nature of *Shq1* precludes constitutive knockout; a conditional/humanized mouse recapitulating the full phenotype is lacking.
5. **GRP78/apoptosis branch** is in-vitro only and its in-vivo contribution to neurodevelopment is uncertain.
6. **Ontology/nomenclature confusion** between *SHQ1* and *KCNN2* "NEDDS" risks misannotation in knowledge bases.
7. **No therapeutic data.** No disease-modifying therapy or trials; treatment evidence is anecdotal (e.g., single-case cannabidiol observation).
8. **Several supporting citation snippets remain "unchecked"** against stored abstracts in the knowledge state and should be verified before final database ingestion.

Proposed Follow-up Experiments / Actions

1. **Establish a conditional/hypomorphic mouse model** (neural-specific *Shq1* reduction or knock-in of recurrent human alleles, e.g., p.Leu333Val) to test whether the ribosomopathy produces cerebellar atrophy, dystonia, and dopaminergic deficit in vivo.
2. **Mechanistically link ribosome biogenesis to the dopaminergic deficit** — e.g., ribosome profiling / selective translational readout in dopaminergic neurons derived from patient iPSCs.
3. **Patient iPSC-derived neurons/organoids** to characterize cortical/cerebellar/dopaminergic differentiation, migration, and ER-stress/apoptosis under patient genotypes.
4. **Genotype–phenotype registry** aggregating all reported cases (allele type vs MRI findings, progression, seizure burden, CSF HVA) to formalize correlations.
5. **Standardize nomenclature/ontology** — ensure MONDO:0859258 ↔ OMIM:619922 ↔ *SHQ1* mapping is propagated and clearly separated from *KCNN2* OMIM:619589 across databases.
6. **Therapeutic exploration** — evaluate agents that enhance residual SHQ1/dyskerin function or boost ribosome biogenesis, and systematically assess dopaminergic/symptomatic therapies (including the anecdotal cannabidiol signal) in prospective, even n-of-1, designs.

7. **Verify all citation snippets** against source abstracts and upgrade "unchecked" evidence items before knowledge-base entry.

Report compiled from 9 confirmed findings and 13 reviewed papers across 5 investigation iterations. Evidence sources span human clinical case series, yeast and mouse model organisms, and in-vitro cellular assays.
